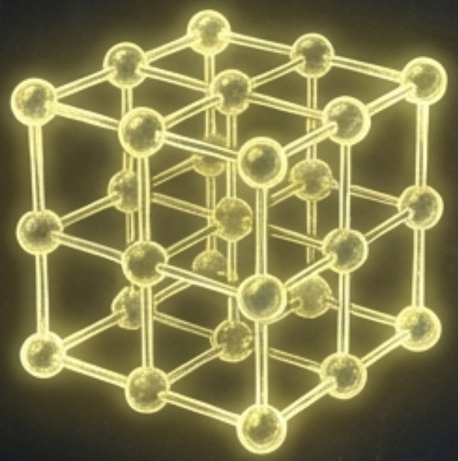
The AI Catalyst

Redefining Research in Applied Chemistry



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The Modern Chemist’s Quest: Architecting Tomorrow’s World



Functional Materials Science

Creating solutions for energy, semiconductors, and catalysis.



High-Efficiency Synthesis

Designing novel biomedical compounds and sustainable green chemistry processes.



Computational Modeling

Integrating data science with laboratory work for predictive insights.

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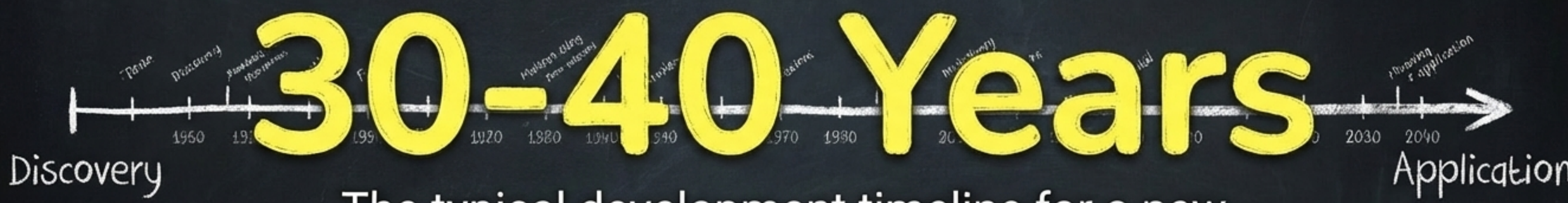
The Great Obstacle: A Process of Trial and Error

The chemist as a methodical, trial-and-error experimenter.

- R&D is a slow, resource-intensive workflow.
- Discovery is often limited by the sheer volume of potential compounds.
- The path from lab concept to industrial production is long and complex.

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The Decades-Long Marathon



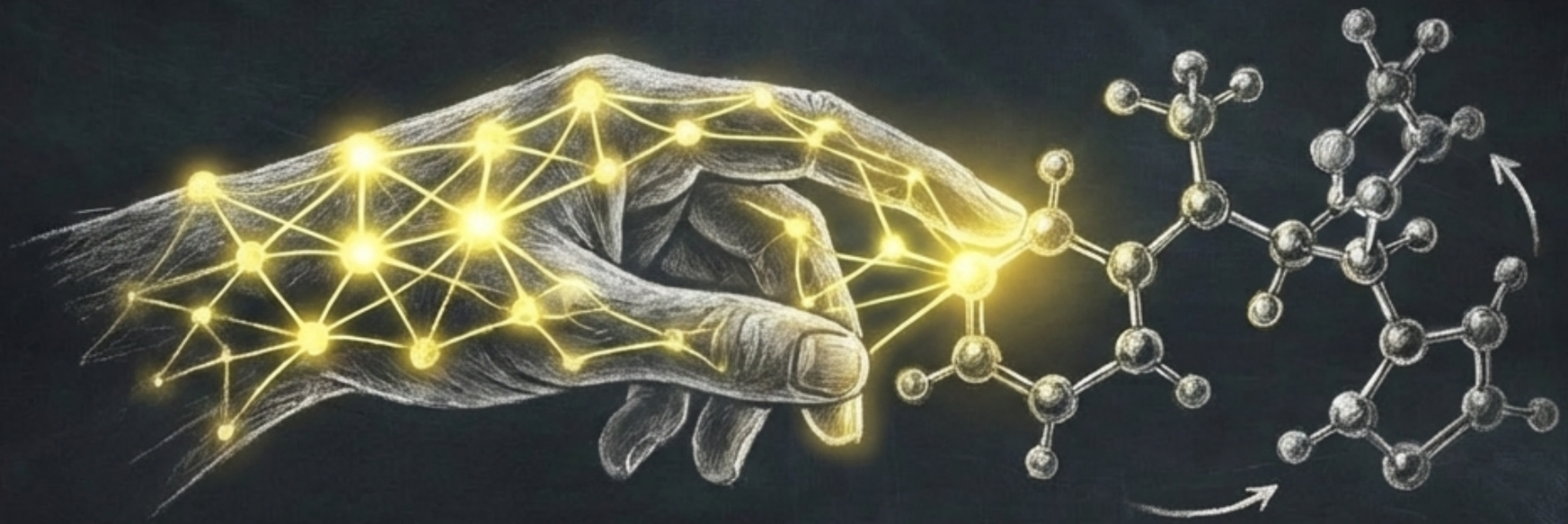
The typical development timeline for a new material, from discovery to market.

This pace is no longer sufficient to meet the speed of technological innovation in sectors like 5G/6G communications or next-generation energy.

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The Emergence of a New Catalyst: Generative AI

A fundamental shift in the R&D paradigm. AI augments the chemist's intuition and accelerates the entire discovery process.



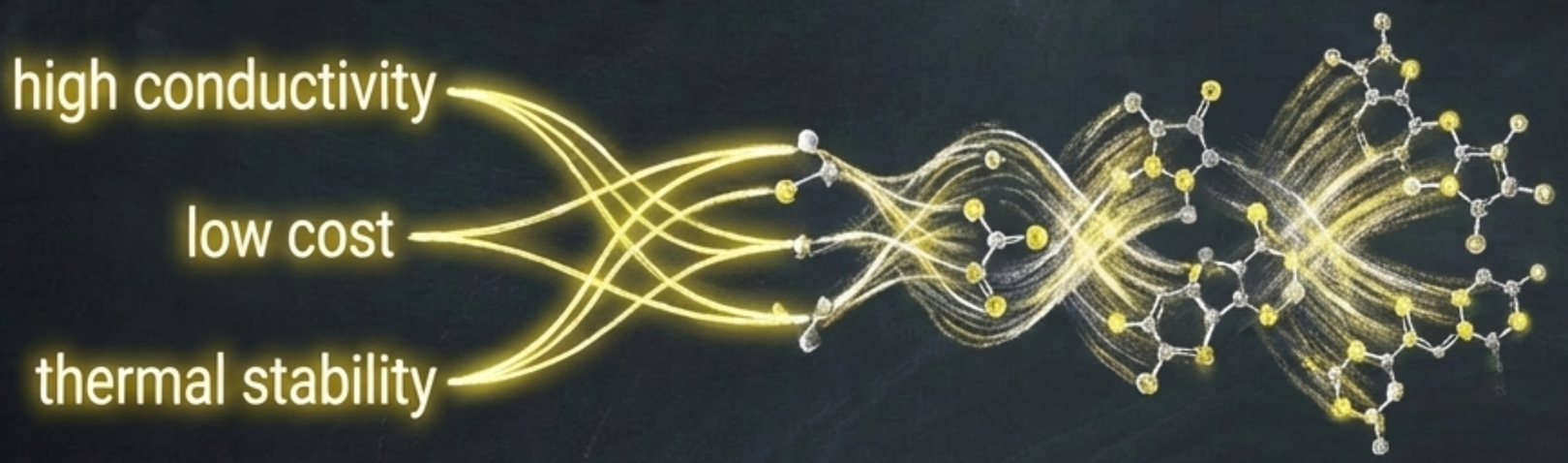
From "Trial-and-Error Experimenter"

To **"Predictive Modeling & Validation Expert"**

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The AI Toolkit: Four Core Capabilities

1. Discovery & Design: Generative Chemistry



AI designs novel molecules and materials with desired properties from scratch.

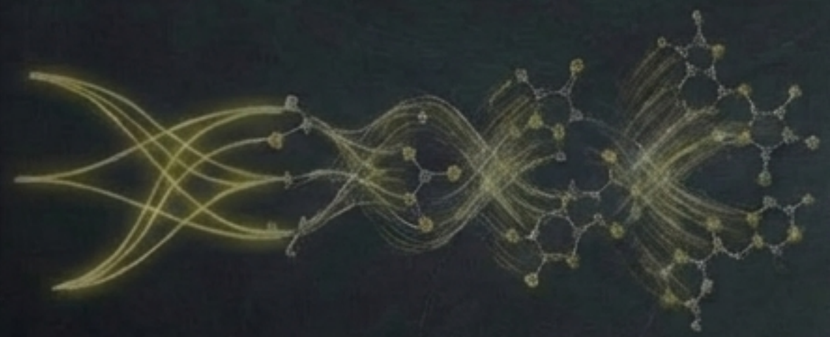
It moves beyond screening existing compounds to direct generation.

The MatterGen model generates materials that are **>2x more likely to be novel and stable.**

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The AI Toolkit: Four Core Capabilities

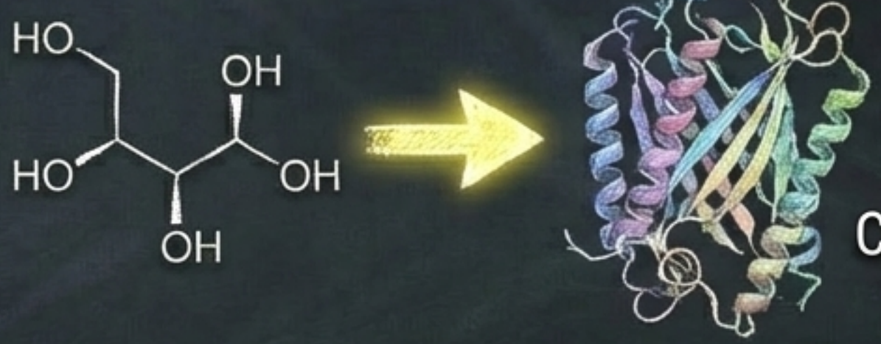
1. Discovery & Design: Generative Chemistry



AI designs novel molecules and materials “from scratch.

It moves beyond screening existing tine compound to direct geraton.

2. Prediction: Properties & Structure



Reactivity: Low ✓

Strength: High ✓

Conductivity: High ✓

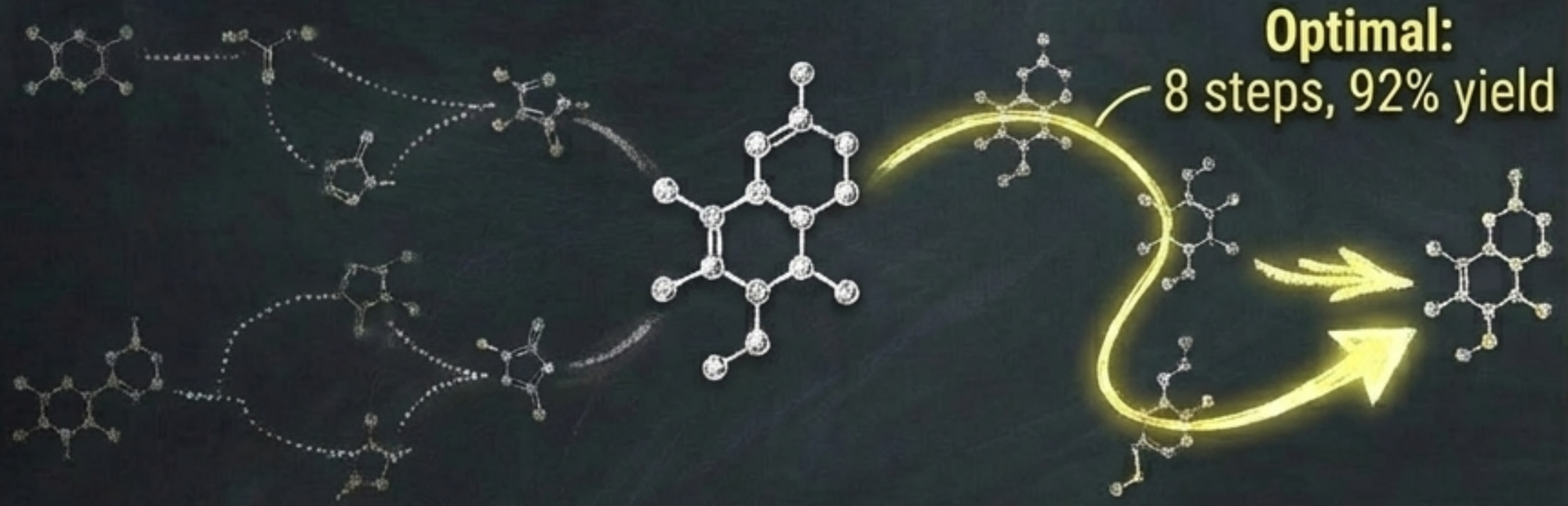
- **Property Prediction**: Forecasts strength, reactivity, and electrical conductivity.
- **Structure Prediction**: Models like AlphaFold enable structure-driven drug design.

An NCKU team used AI to accelerate R&D for **dielectric materials** needed for 5G and 6G technology.

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The AI Toolkit: Four Core Capabilities

3. Optimization: Synthesis & Process Planning



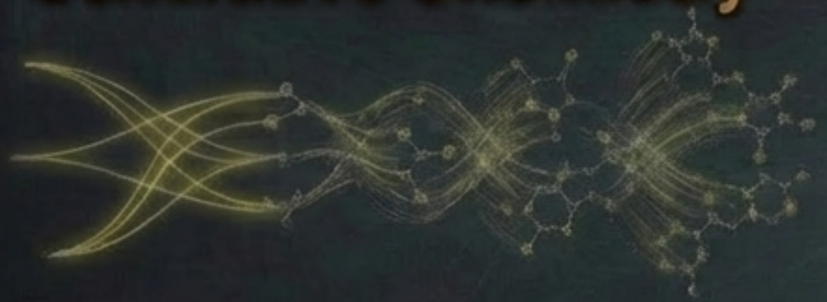
- AI proposes the most efficient chemical synthesis routes.
- It designs a molecule *and* simultaneously outlines the steps to create it.

AI platforms built on extensive materials databases can achieve **>90% accuracy**.

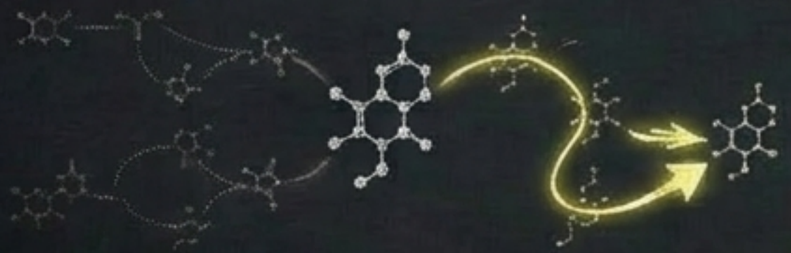
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The AI Toolkit: Four Core Capabilities

1. Discovery & Design: Generative Chemistry



3. Optimization: Synthesis & Process Planning



4. Analysis: Data Management & Literature Insight



- AI models can parse massive volumes of biomedical and chemical literature.
- They extract insights, identify trends, and propose new hypotheses.

The **Human Element**: Underscores the importance of **data literacy**: the ability to manage datasets and critically validate AI-generated insights.

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The New Velocity: From a Marathon to a Sprint



AI transforms materials science from a decades-long process into a rapid-response model capable of keeping pace with technological change.

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The Evolution of the Expert

THE PAST:

The Trial-and-Error Experimenter

Manual, Sequential, Intuition-Based



THE FUTURE:

The Predictive Modeling & Validation Expert

Predictive, Synergistic, Data-Driven

The chemist’s core expertise is not replaced; it is elevated. The role shifts from generating data to directing AI, interpreting its output, and applying critical human judgment.

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The New Frontier of Applied Chemistry

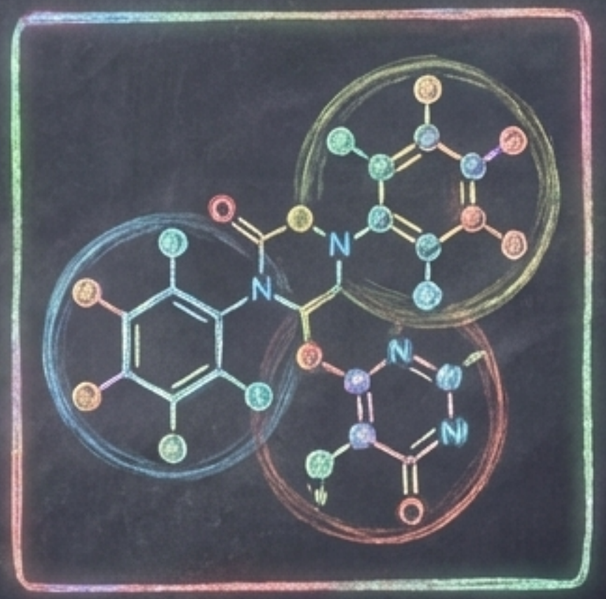
With AI as a catalyst, chemists are now positioned to solve humanity's most pressing challenges at an unprecedented rate.



Energy Solutions: Rapidly designing next-generation solid-state electrolytes and high-efficiency catalysts.



Advanced Semiconductors: Accelerating the development of ultra-pure precursors and materials for future nodes.



Biomedical Breakthroughs: Engineering novel therapeutics and optimizing pharmaceutical synthesis with predictive accuracy.

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